

9 (a) The Venus fly trap, *Dionaea muscipula*, is a carnivorous plant that captures small insects in a trap made from a specialised leaf.

- (i) The first step in capturing an insect occurs when the insect touches at least two sensory hairs on the leaf of the plant within 35 seconds. The second step occurs when an action potential is produced and spreads over the leaf.

Outline how the stimulation of sensory hair cells of a Venus fly trap plant leads to an insect being trapped.

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- (ii) A Venus fly trap can carry out photosynthesis.

Suggest why the Venus fly trap also captures and feeds on insects.

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- (b) The nervous and endocrine systems of a mammal coordinate responses to changes in the internal and external environment.

Table 9.1 compares some features of the nervous and endocrine systems.

Complete Table 9.1.

Table 9.1

feature	nervous system	endocrine system
signal	impulse	
how signal is transmitted		in blood
type of communication	electrical	
typical duration of effect		

[4]

[Total: 9]